

The Luna ecosystem: integrated tools for sleep signal visualization and analysis

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Summary

Sleep physiology generates rich, high-dimensional signals central to understanding brain function, aging, and disease, yet the field has long been constrained by proprietary formats, closed-source algorithms, and limited open, scalable tools spanning both large-cohort analysis and detailed inspection of individual recordings. The Luna ecosystem is an open-source sleep analysis stack built around the Luna C/C++ library, its Python interface (`lunapi`), and Lunascope, an interactive desktop application for visualization, annotation, and review. Scripted and graphical workflows operate on a common data model and command set. Lunascope provides a synchronized multichannel viewer with hypnogram display, spectral summaries, annotation handling, automated sleep staging, cohort-level exploration, multiday actigraphy views, and direct access to the National Sleep Research Resource (NSRR). An embedded scripting console exposes Luna's command language inside the application, allowing users to move between visual inspection and programmatic analysis within a single session.

Statement of need

Computational analysis of polysomnographic (PSG) recordings has become central to sleep research. Large-scale datasets hosted by the NSRR have made thousands of whole-night recordings freely available, enabling population-level studies of sleep architecture, spindle activity, slow oscillations, and EEG-derived biomarkers (Nataliia Kozhemiako, Mylonas, et al., 2022; N. Kozhemiako et al., 2024; Purcell et al., 2017). These analyses increasingly rely on open-source tools that can process recordings at scale, but many sleep researchers, clinicians, and trainees lack fluency with terminal-based or notebook-based workflows, and existing graphical tools for sleep data often emphasize either signal viewing or analysis rather than both within a reproducible open-source workflow.

The Luna/`lunapi`/Lunascope stack was developed to bridge this gap. Luna provides a reusable analytical core, `lunapi` makes that core available in Python for notebooks and batch workflows, and Lunascope adds an interactive environment for review and exploratory analysis. Luna has been used as the primary analytic platform in studies analyzing over 10,000 PSG recordings across multiple cohorts in work by our group (Djonlagic et al., 2021; Nataliia Kozhemiako, Mylonas, et al., 2022; Nataliia Kozhemiako, Wang, et al., 2022; N. Kozhemiako et al., 2024; Purcell et al., 2017) and others (Adra et al., 2022; Hanif et al., 2025; Lokhandwala et al., 2025). Lunascope brings those methods into an interactive graphical workflow.

Luna is also part of the broader US National Heart, Lung, and Blood Institute NSRR ([National Sleep Research Resource](#), 2026; [Zhang et al.](#), 2018) analytical ecosystem: in addition to sharing data, the NSRR has supported the use of common open-source tools for sleep research, and Luna has been used within NSRR analysis workflows. Lunascope includes a Moonbeam module that imports NSRR data directly into the environment used for visualization, review, and downstream analysis.

State of the field

Existing software relevant to PSG analysis reflects different design goals and usage contexts. EDFbrowser ([Beelen](#), 2010) is a mature and widely used viewer, well suited for manual signal inspection, but it is primarily a viewing application rather than an integrated sleep-analysis environment. Broader EEG ecosystems such as EEGLAB ([Delorme & Makeig](#), 2004) and MNE-Python ([Gramfort et al.](#), 2013) are powerful and widely used, but they are general electrophysiology frameworks rather than dedicated PSG review environments. Sleep-focused research tools including YASA ([Vallat & Walker](#), 2021), Wonambi ([Wonambi](#), 2026), SleepTrip ([Cox et al.](#), 2024), the Snooz Toolbox ([Snooz Toolbox](#), 2026), and SleepEEGpy ([Falach et al.](#), 2025) support useful scripted workflows for staging, detection, preprocessing, or artifact cleaning, but they tend to emphasize individual pipelines or high-level wrapper layers rather than a shared analytical engine spanning command-line, Python, and interactive desktop review. Commercial PSG review systems support clinical scoring and reporting, but are typically closed-source, proprietary, and difficult to adapt for reproducible research pipelines or large-scale open-data workflows.

Taken together, existing tools tend to separate signal viewing, algorithmic sleep analysis, and cohort-scale data exploration into different environments. This separation creates friction for reproducible research: visual review, scripted analysis, annotation management, and population-level summaries often occur in disconnected tools with different data assumptions. The Luna ecosystem was developed to reduce this separation by combining a scalable sleep-analysis engine, Python access through lunapi, an interactive desktop application for PSG review and annotation, cohort-level exploration tools, and direct workflows for NSRR-hosted data.

Many existing tools also concentrate on narrower task domains such as staging, visualization, or selected spindle, spectral, or preprocessing pipelines. Luna instead provides a shared analytical engine spanning dataset manipulation, interval-level annotations, multi-day and actigraphy-based analyses, statistical modeling, high-density EEG, and multiple spectral methods. This build-versus-contribute decision reflects the central contribution of the ecosystem: integrating scalable analysis, Python workflows, and interactive review around a common data model and command language.

Software design

The software stack is organized in three layers. Luna, implemented in C/C++, provides the analysis engine and command set for sleep signal processing, including EDF/EDF+ inputs ([European Data Format \(EDF\) and EDF+](#), 2026; [Kemp et al.](#), 1992; [Kemp & Olivan](#), 2003), XML and Luna annotation formats, spectral analyses, automated staging, and statistical modeling. Its native API exposes recordings and derived data to higher-level interfaces, while the same Luna command language is available at the command line and in the application. [Figure 1](#) gives an overview of the core Luna tools, primary functionality, and the visual Lunascope interface.

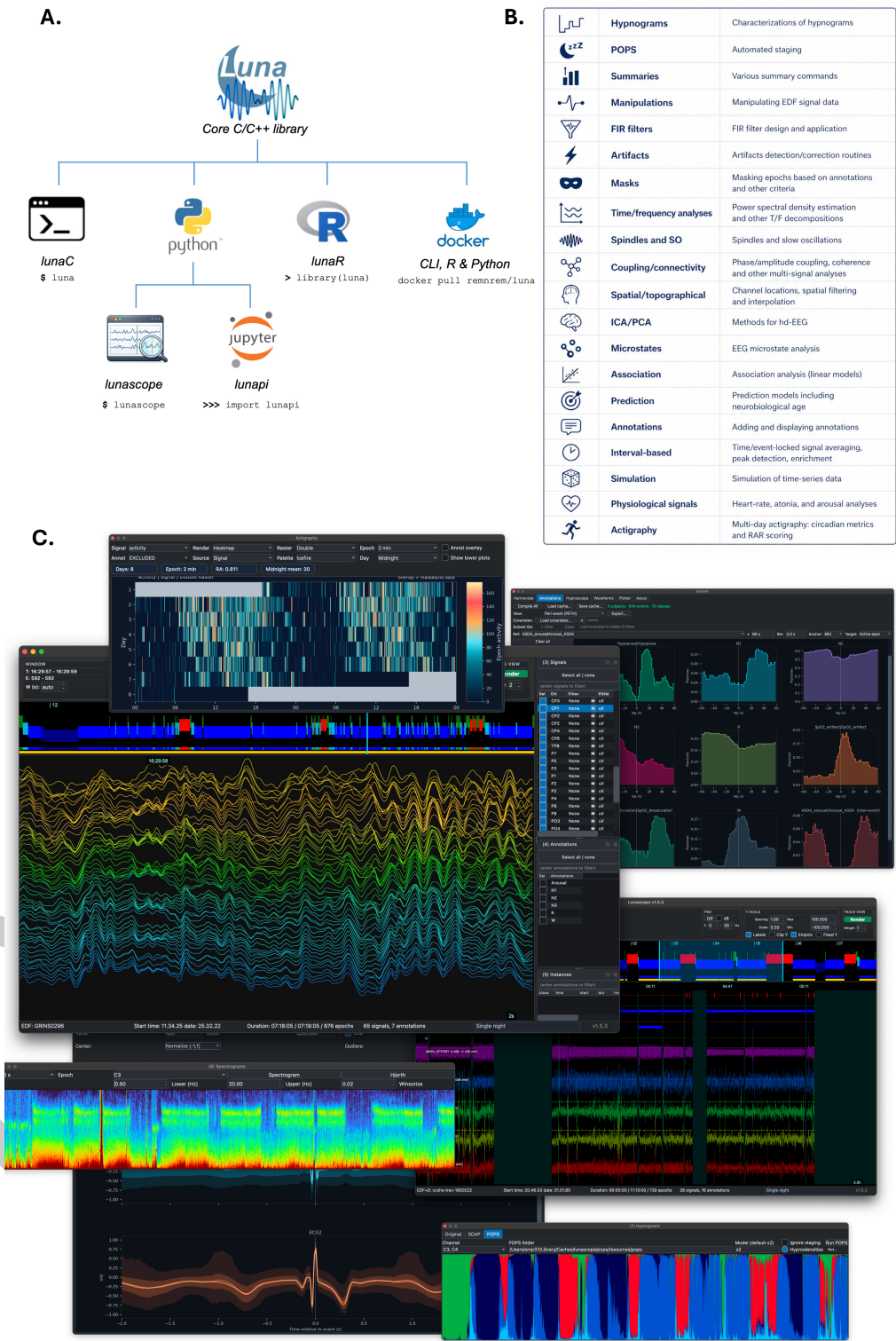


Figure 1: Figure 1. Overview of the Luna & Lunascope. A. The family of Luna tools. B. Schematic of core domains of functionality in Luna. C. Visualization in Lunascope.

87 `lunapi` exposes that engine to Python, allowing the same recording objects and commands
88 to be used in scripts, notebooks, and other applications. It supports single-recording and
89 project-level workflows, including sample-list construction, command execution across cohorts,
90 and structured retrieval of Luna result tables as Python objects.

91 Luna separates computation from presentation: command outputs can be written to result
92 stores, returned to Python as structured objects, or displayed in Lunascope, preserving
93 comparable results across command- line, scripted, and interactive settings.

94 Lunascope is implemented in Python (3.9–3.14) using PySide6 (Qt 6) and pyqtgraph as the
95 graphical layer on top of `lunapi`, meaning every Luna command available at the command
96 line is also available within the desktop application. A native desktop application was chosen
97 over a web-based approach because PSG recordings are typically large, multi-channel, and
98 require responsive pan/zoom interaction across hours of data. `lunapi` provides a lighter-
99 weight embedded viewer (`scope`) for JupyterLab; Lunascope is the more full-featured desktop
100 environment within the same ecosystem.

101 The Lunascope interface is organized around a synchronized signal viewer with docks for
102 annotation, analysis, and cohort exploration. Its key design choice is tight coupling to the
103 Luna data model rather than treating visualization as a passive display layer. Signals and
104 annotations can be inspected or modified in the viewer and are immediately synchronized with
105 commands executed through the console.

106 More generally, interval-level annotations are treated as first-class objects within Luna, and
107 arbitrary expressions can be evaluated against those intervals, their metadata, and the underlying
108 signals. For efficient navigation of long recordings, Lunascope maintains rendered, decimated
109 signal caches and summary envelopes for broad time windows while preserving access to the
110 underlying sample-level data. This supports interactive measurement, annotation-aware review,
111 and direct visual inspection of outputs from Luna analyses.

112 The embedded console uses native Luna commands, while Moonbeam and the Explorer connect
113 NSRR data and cohort-level displays to the same analytic context. This design allows outputs
114 to be inspected as part of the visual workflow rather than only as exported tables, while the
115 command- line tool supports parallel workflows whose results can be combined with `destrat`.

116 Research impact statement

117 The Luna/`lunapi`/Lunascope stack supports a shared workflow spanning batch analysis, scripted
118 exploration, and interactive review. Lunascope is used for PSG review, quality assessment, and
119 exploratory analysis in projects that also rely on Luna's analytical pipeline, including work on
120 sleep EEG spectral variation (Nataliia Kozhemiako, Mylonas, et al., 2022), neurodevelopmental
121 sleep architecture (N. Kozhemiako et al., 2024), and the GRINS schizophrenia neurophysiology
122 consortium (Nataliia Kozhemiako, Wang, et al., 2022; Murphy et al., 2024; Wang et al., 2024).

123 Availability

124 The primary entry point for the Luna ecosystem is <https://zoo-luna.org/luna/>. Lunascope
125 is distributed via PyPI ([Lunascope on PyPI](#), 2026) and as standalone macOS and Windows
126 binaries built via GitHub Actions. The software is available at the Lunascope source repository
127 ([Lunascope Source Repository](#), 2026), with user documentation available separately ([Lunascope](#)
128 [Documentation](#), 2026). Luna documentation is available online ([Luna Documentation](#), 2026),
129 including worked examples, vignettes, and detailed descriptions of command options and outputs.
130 Luna and `lunapi` are also distributed as Docker containers, and `lunapi` is accompanied by
131 interactive notebooks and tutorials ([LunAPI Tutorial and Reference Notebooks](#), 2026). We
132 have also developed an extensive six-part walkthrough ([Luna Walkthrough](#), 2026), built around

20 hd-EEG recordings available through the NSRR Luna/GRINS dataset ([Luna/GRINS Dataset, 2026](#)). The Luna source repository is also publicly available ([Luna Source Repository, 2026](#)).

AI usage disclosure

Generative AI tools (Claude, Anthropic and Codex, OpenAI) were used to assist in editing and formatting the text of this manuscript, and during software development, primarily to help refactor code, generate help-related functions and documentation, and assist with use of the Qt library in Lunascope. All content and code were reviewed, verified, and revised by the authors.

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